

Configure API-KEY

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1. Course Content

- Use the API-KEY registered earlier to configure the robot vehicle's API-KEY.

Warning

Note: Please make sure the vehicle is connected to the internet to use the cloud model services.

2. Start the Dify Service

Tip

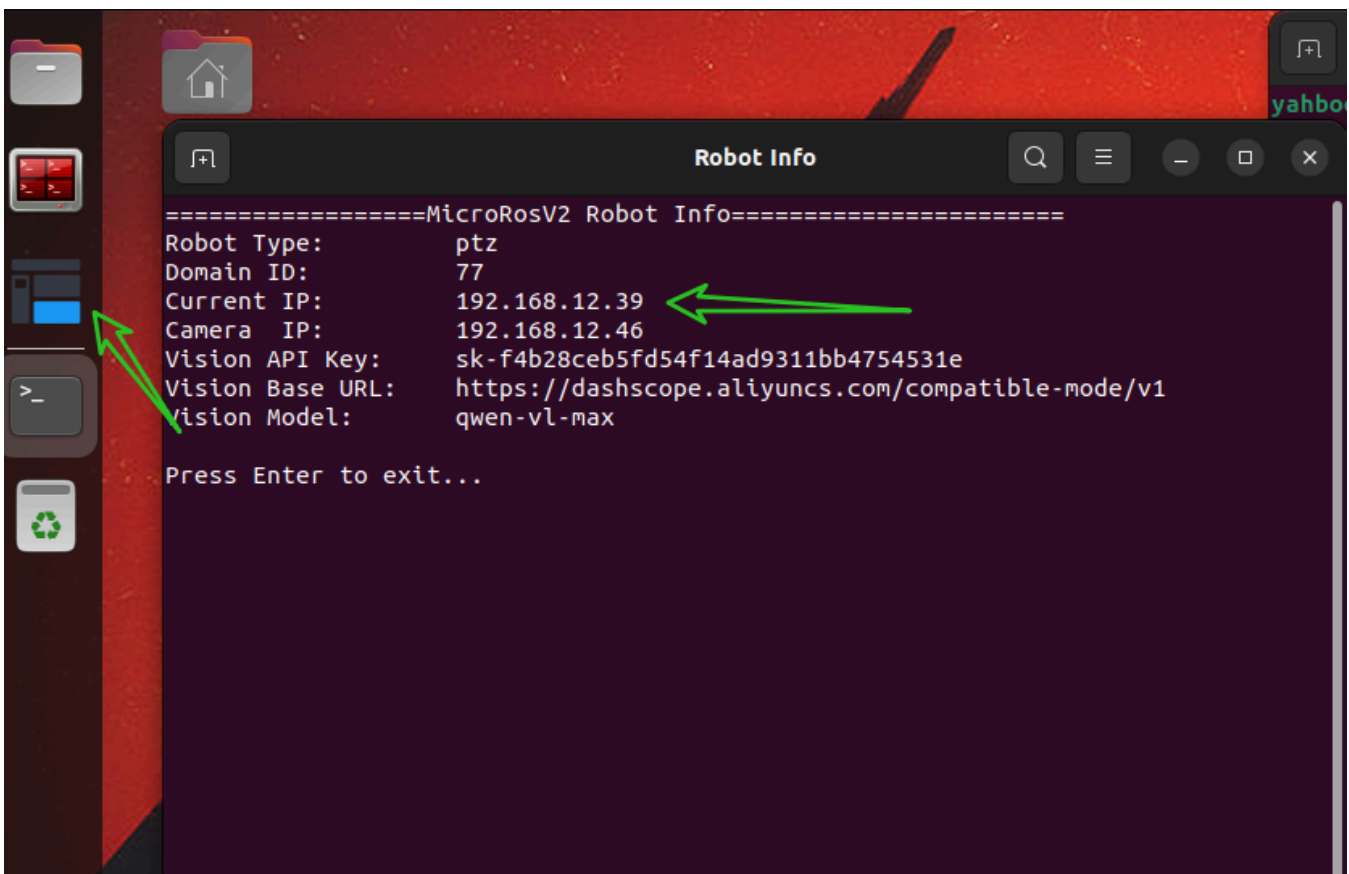
MicrorosV2 builds a multi-agent system through Dify, and Dify centrally manages cloud model calls. Dify starts automatically at boot by default and usually does not need to be started manually. If you need to start it manually, refer to the startup command here.

- Connect to the vehicle via VNC or SSH, and enter the command in the terminal:

```
sh bringup_dify.sh
```

```
jetson@yahboom: ~  
jetson@yahboom: ~ 101x24  
jetson@yahboom:~$ sh bringup_dify.sh  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
WARN[0000] The "CERTBOT_EMAIL" variable is not set. Defaulting to a blank string.  
WARN[0000] The "CERTBOT_DOMAIN" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_USERNAME" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_PASSWORD" variable is not set. Defaulting to a blank string.  
WARN[0000] The "DB_DATABASE" variable is not set. Defaulting to a blank string.  
[+] Running 11/11  
✓ Network docker_ssrf_proxy_network Created 0.0s  
✓ Container docker-db-1 Healthy 2.2s  
✓ Container docker-redis-1 Starting 1.2s  
✓ Container docker-web-1 Started 1.3s  
✓ Container docker_ssrf_proxy-1 S... 1.4s  
✓ Container docker-sandbox-1 Star 1.1s
```

- Check the IP address of the virtual machine by clicking "Robot Information" on the left side of the desktop.

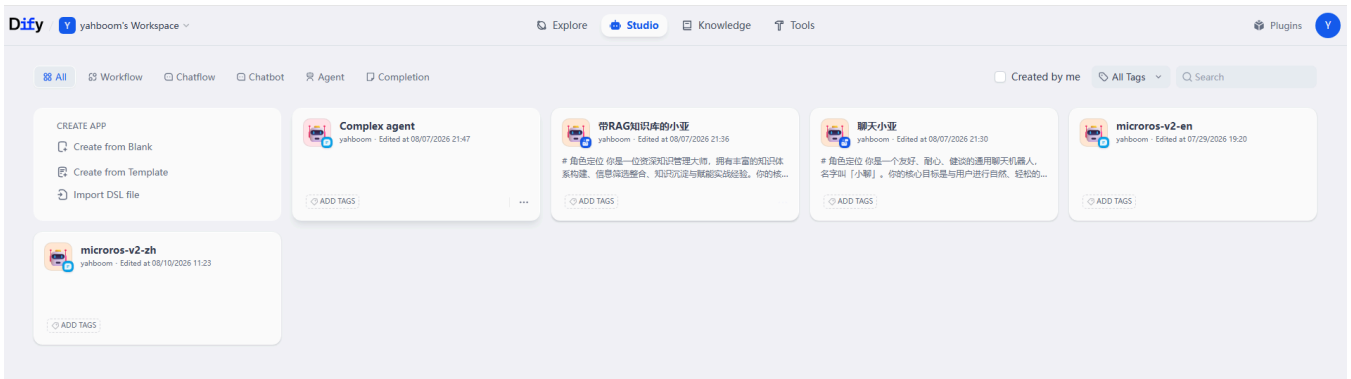


- Enter the vehicle IP address directly in the browser address bar to access the Dify management page. If the device is logging in for the first time, you need to use the account and password. You can select the language in the upper-left corner.

Note

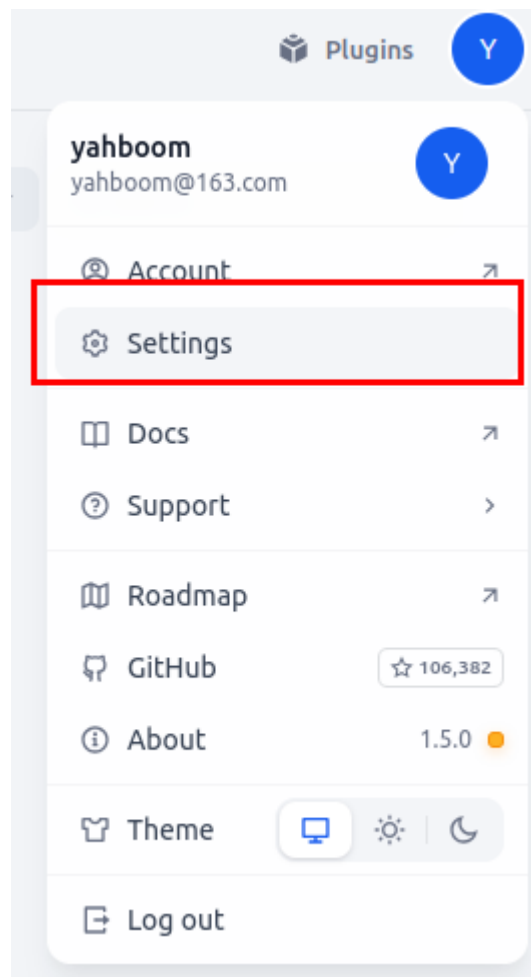
- Account name: yahboom@163.com

- Password: yahboom123
- All account passwords, agent applications, and RAG data are stored locally.
- After entering, it looks like the following figure:

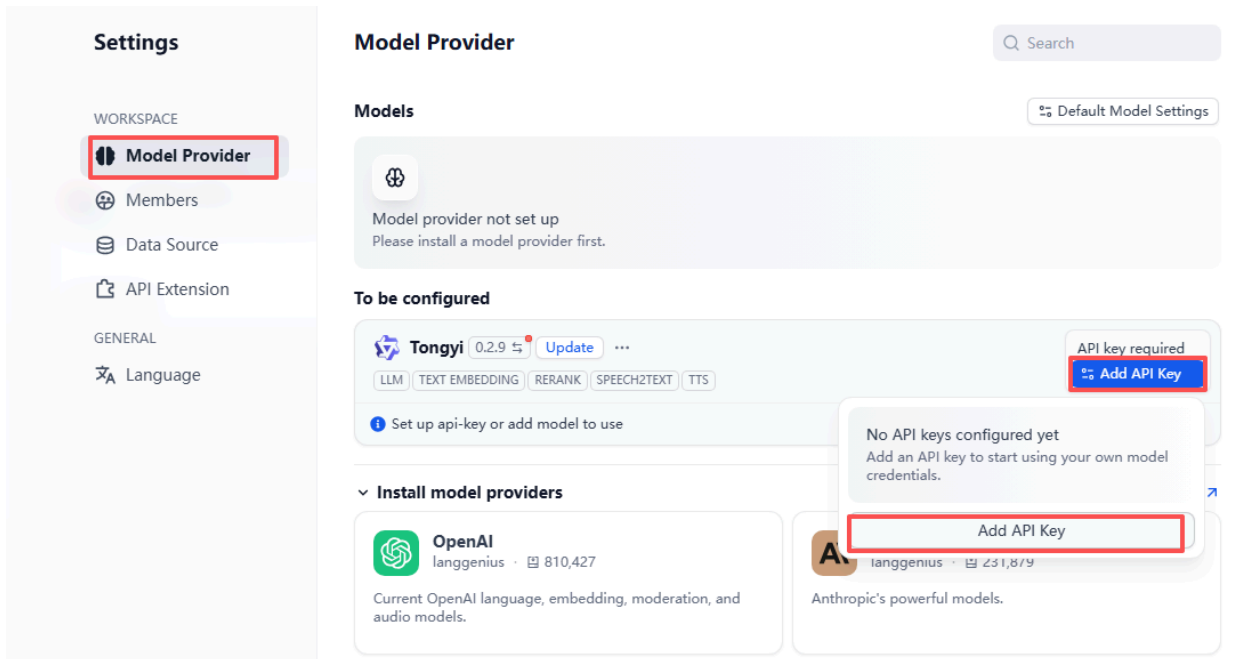


3. Configure Model Provider API-KEY

- Click "Settings".



- Here, configuring the Bailian account API is used as an example. Click "Model Provider" —> "Configure" —> "Add API Key".



- Fill in your Bailian platform API-KEY, then select whether it is an international site account, and click "Save".

API Key Authorization Configuration

After configuring credentials, all members within the workspace can use this model when orchestrating applications.

Authorization Name

Please enter

API Key *

Enter your API Key

Use International Endpoint *

☒ True ☐ False

[Get your API key from AliCloud](#) [Cancel](#) [Save](#)

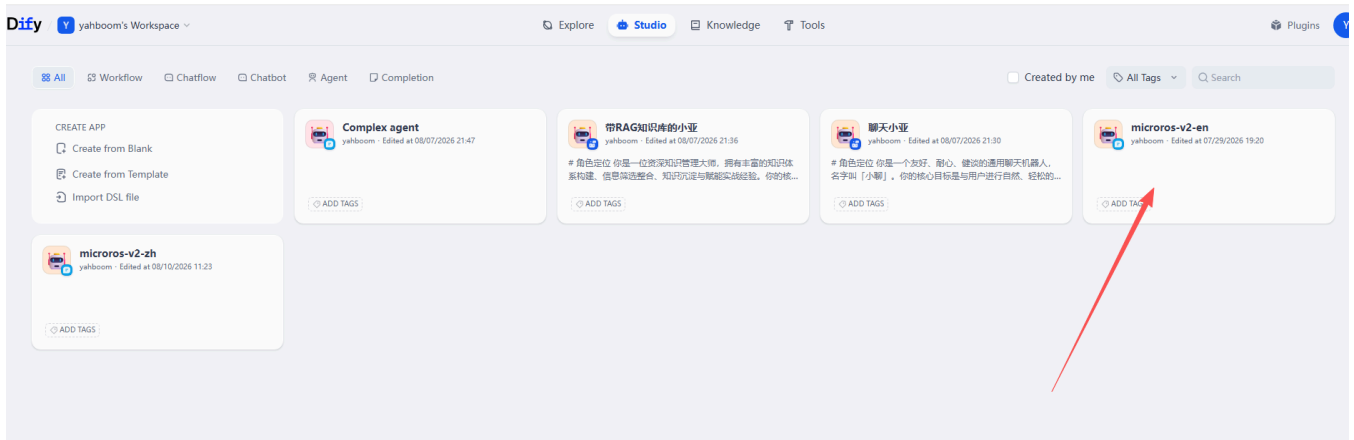
Your API KEY will be encrypted and stored using PKCS1_OAEP technology.

4. Test API-KEY

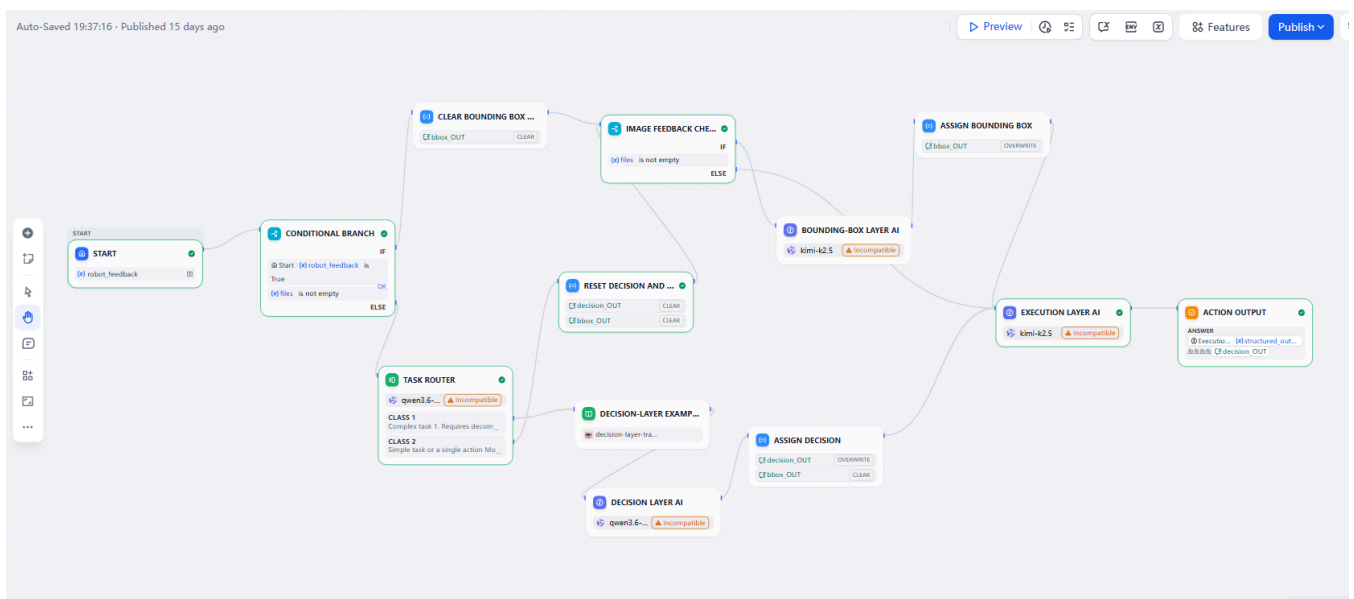
Tip

- If you need to test whether the API-KEY works, refer to this section of the tutorial. If not, you can ignore it.
- Click the microrros-v2-zh application in the Studio (international customers can use the microrros-v2-en

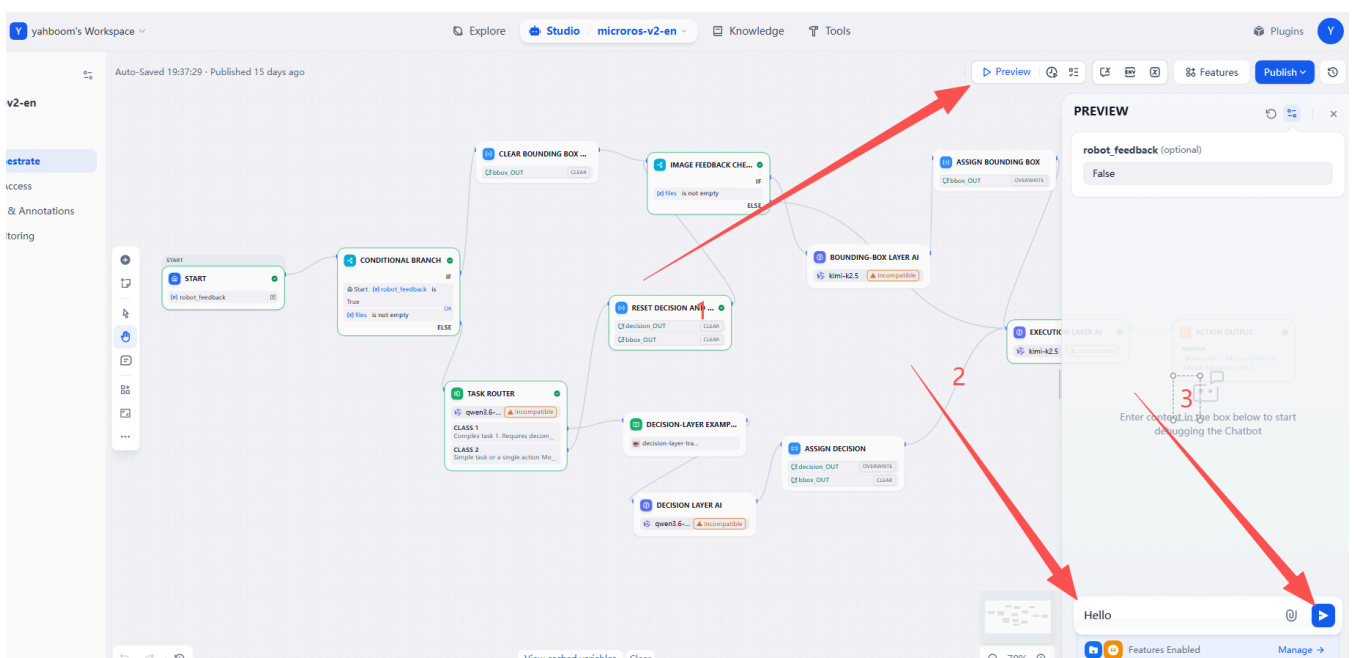
application to test).



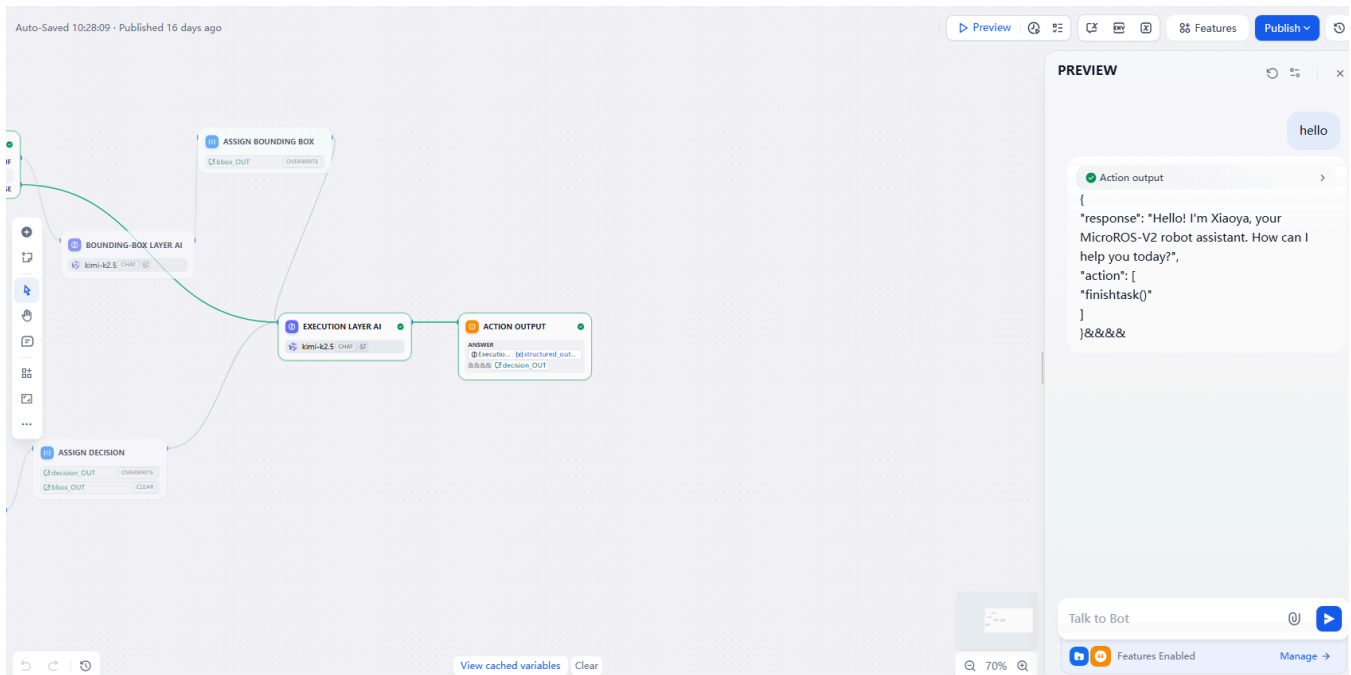
- After entering, you can see the workflow view.



- Click "Preview", then input "hello" and press Enter.



You can see the agent's reply.

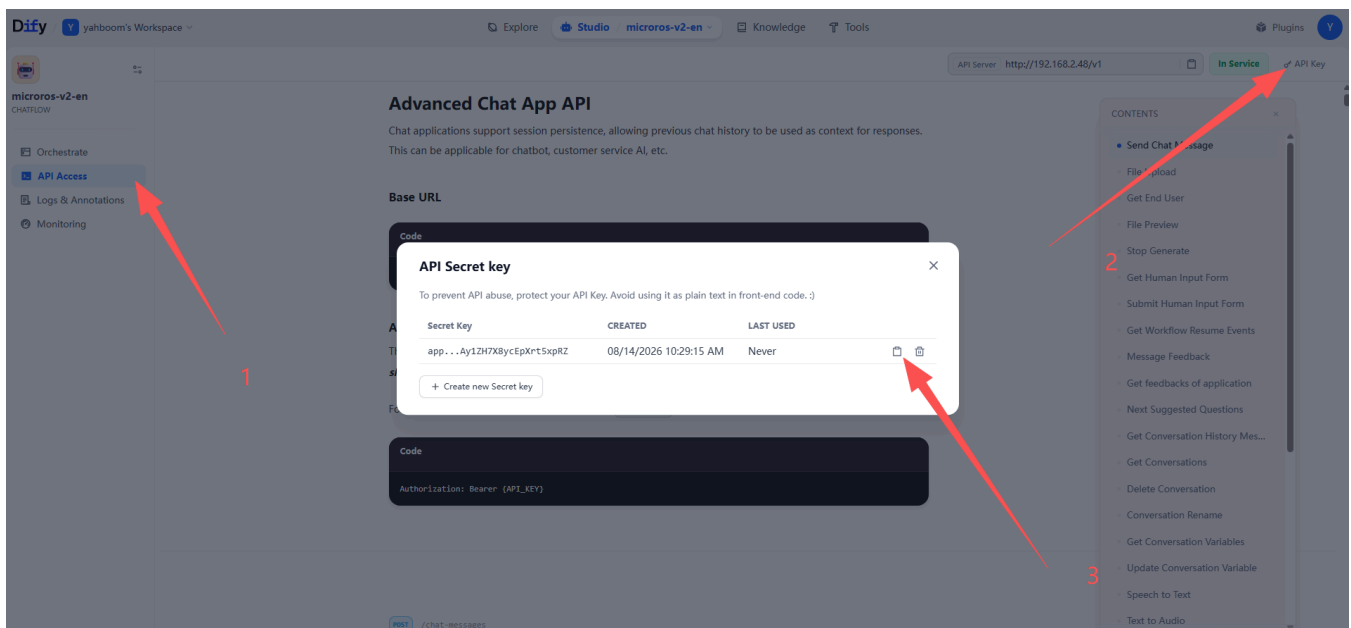


5. Configure the multi_brains Package API

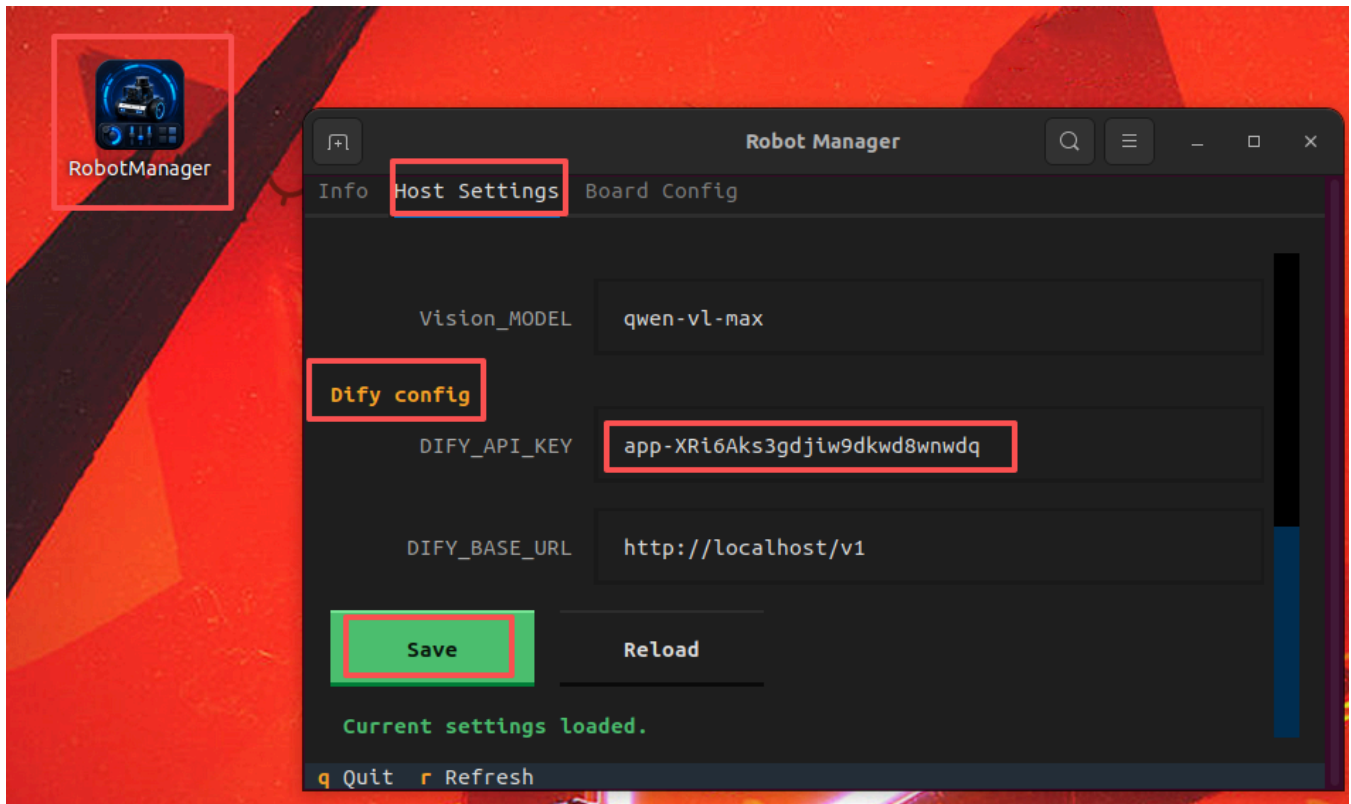
Important

The Dify application key information is configured by default at the factory, so no further configuration is needed. When you need to configure it manually during development, refer to the following content.

- You can find the current Dify key in the Dify backend. `DIFY_API_KEY` is the API access credential used by the multi_brains program to access the Dify application. (Note: This is the API key for microris-v2-en)



- Open the RobotManager application on the desktop, click "Host Settings" — "Dify Configuration" — "DIFY_API_KEY", and paste in the API key copied in the previous step.



6. Parameter File Modification

Important

- When you have development needs to modify the default location files, refer to the following content.
- Parameter file path

```
$HOME/microros_ws/src/multi_brains/config/llm_agent_config.yaml
```

```

src > multi_brains > config > ! llm_agent_config.yaml
1
2  DEBUG_MODE: false # When set to true, prints DIFY/PARSE/ACTION_SERVICE internal details
3  LANGUAGE: "zh"
4
5  #####
6  # Dify Action Feedback Loop
7  #####
8  FEEDBACK_LOOP:
9      # The initial user request is round 1; subsequent action results or image feedback trigger the n
10     max_rounds: 12
11     success_query: "Robot Feedback: {actions} completed"
12     failure_query: "Robot Feedback: {actions} failed"
13     image_query: "Robot image feedback completed"
14     surroundings_query: "Robot multi-directional image feedback completed, image sequence: {directio
15
16     #####
17     # Navigation Configuration
18     #####
19     NAVIGATION:
20         enable: true
21         map_mapping_file: "/home/yahboom/microros_ws/src/multi_brains/config/map_mapping.yaml"
22
23     #####
24     # Line Following Configuration
25     #####
26     FOLLOW_LINE:
27         enable: true
28         # The first four non-empty lines in the HSV file correspond to red, green, blue, and yellow res
29         hsv_file: "/home/yahboom/microros_ws/src/multi_brains/config/color_hsv.txt"
30         finished_topic: "finished"
31         joy_state_topic: "/JoyState"
32         scan_topic: "/scan"
33         buzzer_topic: "/Buzzer"
34         image_width: 640
35         image_height: 480
36         linear: 0.2
37         Kp: 80
38         Ki: 0
39         Kd: 20
40         scale: 1000
41         max_angular_z: 1.5
42         center_tolerance_px: 40
43         line_error_divisor: 16.0
44         min_line_radius: 10.0
45         lost_timeout_sec: 10.0
46         lost_log_after_sec: 5.0
47         control_interval_sec: 0.025
48         camera_start_timeout_sec: 3.0

```

- Other parameter modifications

The `LANGUAGE` parameter: domestic users select `zh`, international users select `en`.

The `DEBUG_MODE` parameter: debug mode. When enabled, you can see logs such as Dify requests, action parsing, and action execution in the terminal.

The `FEEDBACK_LOOP.max_rounds` parameter: limits how many rounds of "Dify decision -> action execution -> result feedback" loop a single user instruction can perform at most, avoiding infinite loops in abnormal situations.

The `NAVIGATION.map_mapping_file` parameter: configures the map point mapping file path.

The `FOLLOW_LINE.hsv_file` parameter: configures the HSV calibration file path read for color line following.

The `CAMERA.image_topic` parameter: configures the camera image topic, defaulting to `/camera/image_raw`.

The current package uses `terminal_client.py` by default for text interaction and no longer configures parameters related to speech recognition, voice wake-up, or speech synthesis. User instructions are sent to `llm_agent_node` via `/llm_text_input`, and the robot's replies are returned to the terminal via `/llm_text_output`.

- After modifying and saving, compile the current `multi_brains` package.
- One-Click Modify and Build Command for International Users

```
sed -i 's/LANGUAGE: "zh"/LANGUAGE: "en"/g'
~/microros_ws/src/multi_brains/config/llm_agent_config.yaml && cd ~/microros_ws && colcon
build --packages-select multi_brains && source install/setup.bash
```

6. Start Text Interaction

- Start the agent core node:

```
ros2 launch multi_brains llm_agent_control.launch.py
```

- Open another terminal and start the text client:

```
ros2 run multi_brains terminal_client
```

- After the terminal client starts, you can directly input natural language instructions. Common control commands are as follows:

```
reset  Reset the Dify conversation context
clear  Clear the screen
help   Display help
exit   Exit the terminal client
```